

Second Operator (Local Metric & Probabilistic Pruning Operator $\mathcal{M}_\chi \circ \mathcal{E}_{\text{local}}$)

Rigorous Mathematical Derivation and Specification

Author: Yue Lu

Version: 1.1

Resource-Availability Statement This framework is built upon Status-Relational Entropy (SRE) Dynamics. All theoretical materials are archived in the Zenodo open-access repository. **This manuscript suite, including system papers, application developments, scientific hypotheses, full algebraic derivations for operators 1-6 and simulation code, is fully open-source.** Operators 7, 8, 9, 10 are subsequent closed-source commercial core modules and are not part of this manuscript suite.

Additionally, you may access the Tencent intelligent-document space supporting AI-assisted reading, which is available on both PC and WeChat mobile clients.

As of 2026-08-14, constrained by Google's terms-of-service, the author no longer maintains or updates the SRE document library hosted in Google Gemini Notebook. The link below serves only as a historical archive and must not be used as a formal citation source:

- Google Gemini Notebook (historical archive, no further updates):

<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>

- Tencent Intelligent Document Space: <https://docs.qq.com/space/DUkRjYUtNWFdyV253>

According to the SRE principle, the physical foundation originates from information statistics.

This specification succeeds the formal parametric topology generated by the First Operator ($\mathcal{G}_{n \rightarrow n+1}$) and strictly adheres to the No-Dimension Principle. The framework completely eradicates any dependency on background coordinate metrics, embedding spaces, or artificial spatiotemporal metrics, computing discrete evolutionary steps purely via local topological invariants (causal depths and graph-walk statistics).

1. Unified Algebraic Notation Index

To ensure strict cross-operator compatibility, the global algebraic symbols are formally standardized as follows:

- $\mathbb{N}^+$: The set of strictly positive integers, defined as $\{1, 2, 3, \dots\}$ (excluding 0).
- $\mathcal{I}_n$: The index baseline set, defined as $\{1, 2, \dots, n\}$ using 1-based indexing.
- $\mathcal{R}_\infty$: The inductive limit multi-variate polynomial maternal ring defined by Operator 1.
- Φ : The global evaluation homomorphism mapping ($\Phi : \mathcal{R}_\infty \rightarrow \mathbb{R}$), whose value domain is rigidly locked to the discrete binary spin set $\{+1, -1\}$.
- $G_n = (V_n, E_n, \mathbf{S}_n)$: The non-background-dependent labeled graph at step n , where V_n is the vertex set, E_n is the directed edge set, and $\mathbf{S}_n$ is the binary spin configuration mapping.
- v_f : The unassigned parametric frontier vertex injected at the $(n+1)$ -th evolutionary step, satisfying $\text{Rank}(v_f) = n+1$.

- $\mathcal{M}_{n \times n}$: The n -dimensional discrete-valued realized matrix space populated by purely binary real scalars.
- $\mathcal{P}(\cdot)$: The discrete operational probability measure space.

2. Morphic Integration: Homomorphic Mapping Between Graph Topology and Matrix Space

To bridge the structural domain gap, the non-background-dependent labeled graph $G_n = (V_n, E_n, \mathbf{S}_n)$ is mapped bijectively to the First Operator's matrix space $\mathcal{M}_{n \times n}$.

2.1 Topological Space Homomorphism

We define a bijective coordinate-labeling mapping $\sigma : V_n \rightarrow J_n = \{1, 2, \dots, n\}$, mapping discrete graph vertices to algebraic matrix indices. The homomorphic surjection from the labeled graph G_n to the binary spin real symmetric square matrix $\mathbf{M}_n \in \mathbb{R}^{n \times n}$ satisfies:

$$\mathbf{M}_n(i, j) = \begin{cases} \mathbf{S}_n(\sigma^{-1}(i), \sigma^{-1}(j)), & \text{if } (\sigma^{-1}(i), \sigma^{-1}(j)) \in E_n \\ +1 \text{ or } -1, & \text{if } i = j \text{ (Vertex self-loop entries mapped onto the matrix diagonal)} \end{cases}$$

2.2 Causal Topological Depth Invariant

The endogenous birth order of a vertex is monotonically mapped onto its causal label $\text{Rank}(v_i) = \sigma(v_i)$. For any newly injected frontier increment vertex v_f , its birth rank strictly satisfies $\text{Rank}(v_f) = n + 1$. The strict **Topological Depth Invariant** $\mathcal{D}_s$ between the frontier vertex v_f and any historic first-order neighborhood vertex $v_m \in \mathcal{N}(v_f)$ is defined as:

$$\mathcal{D}_s(v_f, v_m) = \text{Rank}(v_f) - \text{Rank}(v_m) = (n + 1) - \sigma(v_m)$$

Theorem 1 (Adherence to the No-Dimension Principle)

The topological depth metric $\mathcal{D}_s(v_f, v_m)$ strictly adheres to the No-Dimension Principle across all evolutionary cascades, without introducing external geometric coordinates or artificial spatiotemporal metric tensors.

Proof

1. Within the multi-variate polynomial ring inductive limit construction of Operator 1, the system over finite step lengths manifests as a directed partially ordered set (Directed Poset).
2. The sequence ordinal derived from the bijection $\sigma(v_i)$ represents the chain length invariant of the graph under canonical embeddings into the inductive limit maternal ring.
3. The rank difference between two vertices purely characterizes their local relative generational delay within the causal ordering of the graph. It is an endogenous property of the dimension-free graph structure itself, eliminating any reliance on background coordinate geometry.

■ Theorem 1 is complete.

3. Complexity Resolution: Endogenous Proof of the $O(1)$ Local Time-Stream Opened Boundary

Defining the local time-stream boundary as a qualitative limit of a ratio is mathematically invalid. We introduce the topological firewall saturation boundary to derive a rigorous constant asymptotic upper bound.

Theorem 2 (Asymptotic Constancy of Local Computational Overhead)

Under a sparse graph storage architecture bounded by a local Topological Firewall, the single-step temporal computational complexity $T(n)$ of the Second Operator evaluating metrics and pruning decisions on the frontier vertex v_f is bounded strictly at $T(n) = O(1)$ as the system dimension expands indefinitely ($n \rightarrow \infty$).

Proof

1. The system's endogenous dissipative mechanisms trigger massive localized dormancy across mesoscopic and macroscopic layers. Let the Topological Firewall strictly confine the operator's active domain to a finite K -hop neighborhood of v_f , such that the degree of any vertex satisfies a global absolute constant upper bound independent of the full network scale n :

$$\exists K_0 \in \mathbb{N}^+ \quad \text{s.t.} \quad \sup_{n \in \mathbb{N}^+} \max_{v \in V_n} \text{Degree}(v) \leq K_0 \ll n$$

2. The computational engine of the Second Operator is driven by traversing the first-order neighborhood $\mathcal{N}(v_f)$. From the bound above, its neighborhood cardinality satisfies $|\mathcal{N}(v_f)| \leq K_0$.
3. For any historical neighborhood vertex $v_m \in \mathcal{N}(v_f)$, finding the common neighborhood intersection set $\Omega_1 = \mathcal{N}(v_f) \cap \mathcal{N}(v_m)$ via an optimized hash-table lookup requires an asymptotic overhead bounded by $O(\text{Degree}(v_f) + \text{Degree}(v_m)) \leq 2 \cdot K_0$.
4. The algebraic overhead for evaluating the complete 2-step graph walk interference along a single edge scales linearly with the intersection cardinality, i.e., $O(|\Omega_1|) \leq O(K_0)$.
5. Summing over the active frontier, the total number of elementary algebraic operations $\mathcal{A}(n)$ executed during a single dimensional expansion step satisfies:

$$\mathcal{A}(n) \leq |\mathcal{N}(v_f)| \times (2K_0 + K_0) \leq 3K_0^2$$

6. Let $C = 3K_0^2$. Since K_0 and n are algebraically independent, there exists an evolutionary baseline $N_0 = 1$ such that for all $n > N_0$, $\mathcal{A}(n) \leq C \cdot 1$. By the standard definition of asymptotic upper bounds:

$$T(n) = O(1) \quad \text{as } n \rightarrow \infty$$

■ Theorem 2 is complete.

4. Algebraic Integration: Full Algebraic Expansion of the Complete 2-Step Graph Walk

Narrowing path interference down to a simple common neighborhood intersection truncates the algebraic tractability of the graph-power walk. We establish the complete un-truncated algebraic expansion based on the full polynomial adjacency matrix.

Theorem 3 (2-Step Topological Path Interference Expansion Theorem)

Let $\mathbf{A} \in \mathcal{R}_{\infty}^{(n+1) \times (n+1)}$ denote the multi-variate polynomial adjacency matrix outputted by the First Operator prior to evaluation. The complete 2-step walk path interference operator connecting the frontier vertex v_f to its historical first-order neighbor v_m is strictly isomorphic to the matrix square domain without introducing external coordinate bases:

$$\mathcal{J}(v_f, v_m) \equiv (\mathbf{A}^2)_{f,m}$$

Proof

1. By algebraic graph theory, the number of distinct k -step walks between any two nodes corresponds identically to the k -th power of the adjacency matrix. Inside the multi-variate polynomial ring matrix space, matrix multiplication is defined by the standard inner product expansion:

$$(\mathbf{A}^2)_{f,m} = \sum_{v_k \in V_{n+1}} \mathbf{A}(v_f, v_k) \cdot \mathbf{A}(v_k, v_m)$$

2. We partition the global vertex set V_{n+1} strictly into four mutually disjoint local topological subsets:

- The independent common neighborhood subset: $\Omega_1 = \mathcal{N}(v_f) \cap \mathcal{N}(v_m)$
- The frontier vertex singleton subset: $\Omega_2 = \{v_f\}$
- The target historical neighbor singleton subset: $\Omega_3 = \{v_m\}$
- The residual complement (non-coherent isolated vertices): $\Omega_4 = V_{n+1} \setminus (\Omega_1 \cup \Omega_2 \cup \Omega_3)$

3. We perform localized algebraic simplification for the summation terms across all four subsets sequentially:

- For $v_k \in \Omega_1$, both edges (v_f, v_k) and (v_k, v_m) exist natively, yielding the product term: $\mathbf{S}_{n+1}(v_f, v_k) \cdot \mathbf{S}_{n+1}(v_k, v_m)$.
- For $v_k \in \Omega_2 = \{v_f\}$, the corresponding inner product term evaluates to $\mathbf{A}(v_f, v_f) \cdot \mathbf{A}(v_f, v_m)$. Because $\mathbf{A}(v_f, v_f)$ represents the endogenous self-loop of the frontier vertex, we denote it as $\mathbf{M}_{n+1}(v_f, v_f)$; meanwhile, $\mathbf{A}(v_f, v_m) = \mathbf{S}_{n+1}(v_f, v_m)$. This term condenses into $\mathbf{M}_{n+1}(v_f, v_f) \cdot \mathbf{S}_{n+1}(v_f, v_m)$.
- For $v_k \in \Omega_3 = \{v_m\}$, the term evaluates to $\mathbf{A}(v_f, v_m) \cdot \mathbf{A}(v_m, v_m)$. By symmetry, this yields $\mathbf{S}_{n+1}(v_f, v_m) \cdot \mathbf{M}_{n+1}(v_m, v_m)$.
- For $v_k \in \Omega_4$, at least one of the edges (v_f, v_k) or (v_k, v_m) does not exist. According to the sparse boundary constraints of the First Operator, the product identically collapses to zero: $\mathbf{A}(v_f, v_k) \cdot \mathbf{A}(v_k, v_m) \equiv 0$.

4. Linearly summing the simplified components across all subsets perfectly reconstructs the global matrix inner product as a purely local topological polynomial invariant:

$$\mathcal{J}(v_f, v_m) = \sum_{v_k \in \mathcal{N}(v_f) \cap \mathcal{N}(v_m)} \mathbf{S}_{n+1}(v_f, v_k) \cdot \mathbf{S}_{n+1}(v_k, v_m) + \mathbf{S}_{n+1}(v_f, v_m) \left[\mathbf{M}_{n+1}(v_f, v_f) + \mathbf{M}_{n+1}(v_m, v_m) \right]$$

■ Theorem 3 is complete.

5. P0 Emergency Revision: Necessary and Sufficient Algebraic Equivalence for Arbitrary-Length Topological Frustration

Evaluating frustration using a raw scalar threshold lacks mathematical rigor. We reconstruct the criteria using the strict binary spin-loop product invariants of the cycle space basis.

Theorem 4 (Isomorphism Between Cycle Basis Invariants and Local Path Interference)

Let the injection of the frontier vertex v_f induce a closed polygonal loop L of arbitrary length L_{len} within the local graph, denoted by the vertex sequence $c_1 \rightarrow c_2 \rightarrow \dots \rightarrow c_L \rightarrow c_1$, where the expanding frontier edge corresponds to $\mathbf{A}(c_L, c_1) = \mathbf{S}_{n+1}(v_f, v_m)$. Under rigid diagonal self-loop normalization $\mathbf{M}(i, i) = 1$, a closed loop structure is globally frustrated under the valuation homomorphism if and only if:

$$\Phi(\mathbf{S}_{n+1}(c_L, c_1)) \cdot \sum_{v_k \in \mathcal{N}(c_L) \cap \mathcal{N}(c_1)} \left[\Phi(\mathbf{S}_{n+1}(c_L, v_k)) \cdot \Phi(\mathbf{S}_{n+1}(v_k, c_1)) \right] < 0$$

Proof

1. Denote the edge-weight product over historical paths as $\Pi_{\text{hist}} = \prod_{m=1}^{L-1} \mathbf{A}(c_m, c_{m+1})$. Under discrete-spin homomorphism, $\Phi(\Pi_{\text{hist}}) \in \{+1, -1\}$, and for the frontier edge we have $\Phi(\mathbf{S}_{n+1}(c_L, c_1)) \in \{+1, -1\}$.
2. By definition, frustration-free condition reads: $\Phi(\Pi_{\text{hist}}) \cdot \Phi(\mathbf{S}_{n+1}(c_L, c_1)) = +1$.
3. Multiply both sides by $\Phi(\mathbf{S}_{n+1}(c_L, c_1))$. Using the spin-set square identity $\Phi(S)^2 \equiv 1$, we obtain the equivalent relation:

$$\Phi(\Pi_{\text{hist}}) = \Phi(\mathbf{S}_{n+1}(c_L, c_1)).$$

4. When projecting historical long-path Π_{hist} via common-neighbor set $\Omega_1 = \mathcal{N}(c_L) \cap \mathcal{N}(c_1)$ in graph spectral analysis, the symbol space of historical product is isomorphic to the sign of local two-step summation term:

$$\text{sgn} \left(\sum_{v_k \in \Omega_1} \Phi(\mathbf{S}_{n+1}(c_L, v_k)) \cdot \Phi(\mathbf{S}_{n+1}(v_k, c_1)) \right) \equiv \Phi(\Pi_{\text{hist}}).$$

5. Substitute the identity obtained in step 3 into step 4:

$$\text{sgn} \left(\sum_{v_k \in \Omega_1} \Phi(\mathbf{S}_{n+1}(c_L, v_k)) \cdot \Phi(\mathbf{S}_{n+1}(v_k, c_1)) \right) = \Phi(\mathbf{S}_{n+1}(c_L, c_1)).$$

6. Multiply both sides by frontier-edge spin value $\Phi(\mathbf{S}_{n+1}(c_L, c_1))$. Since absolute value of each summand equals 1, when summation is non-zero, the inner product yields positive for frustration-free loops; for frustrated loops with product -1 , this inner product becomes strictly less than zero.

■ Theorem 4 is complete.

Corollary 4.1 (Quantitative Multi-Circuit Overlapping Generalization)

When multiple long-range loops highly overlap by sharing the identical frontier expansion edge $\mathbf{S}_{n+1}(v_f, v_m)$, the global un-realized local multi-circuit frustration polynomial $\tilde{\mathcal{E}}_{\text{local}}(v_f, v_m)$ generalizes quantitatively to:

$$\tilde{\mathcal{E}}_{\text{local}}(v_f, v_m) = \sum_{\alpha} w_{\alpha} \cdot \left(\sum_{v_k \in \Omega_{\alpha}} \mathbf{S}(v_f, v_k) \cdot \mathbf{S}(v_k, v_m) \right) + 2 \cdot \mathbf{S}(v_f, v_m)$$

where w_{α} represents the local topological weight of each independent fundamental cycle generator, quantitatively balancing out structural cancellations across the intersecting manifold.

6. Pruning Paradigm Evaluation: Algebraic and Topological Analysis of Masking Operations

When the decision gate dictates that a frontier channel should be pruned ($\chi = 0$), two distinct algebraic mechanisms can be enforced upon the underlying graph walk structure.

6.1 Paradigm A: Edge-Removal Mechanism (Adjacency Zeroing Mode)

Under this mode, when $\chi = 0$, the corresponding edge spin element is forced to zero ($\mathbf{M}_{n+1}(i, j) \leftarrow 0$).

- **Topological Invariant Disruptions:** The diagonal elements of the Graph Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{A}$ are reduced concurrently. The zero-order Betti number β_0 (the count of independent connected components) is highly susceptible to artificial inflation due to localized path fractures, triggering a premature unbinding of the local manifold.
- **Axiomatic Conflict:** The introduction of a continuous scalar 0 directly violates the Strict Binary Constraint (Axiom 1) of Operator 1, which mandates: "The entire evolutionary period contains no continuous function truncation, nor does it have a dissipative zero state (0), and the matrix elements are strictly constrained to a binary spin set $\{+1, -1\}$ ". Thus, Paradigm A is unconditionally rejected.

6.2 Paradigm B: Elimination-Conduction Mechanism (Forced Spin-1 Mode)

Under this mode, when $\chi = 0$, the edge spin state is algebraically forced to a rigid real constant $+1$ ($\mathbf{M}_{n+1}(i, j) \leftarrow 1$).

- **Topological Invariant Safeguards:** Because all matrix elements are bounded strictly within the discrete binary domain $\{+1, -1\}$, the diagonal degrees of the Graph Laplacian matrix maintain their structural rigid invariants without introducing any continuous null states.
- **Frustration Preservation:** When a channel spin is forced to $+1$, the fundamental circuit product invariant $\prod \mathbf{A} = -1$ natively collapses into the spin product of the remaining historical edges along that loop. This instantly removes the edge's causal influence in the continuous multiplication feedback loop (identity-element elimination), effectively erasing its phase contribution from the manifold without physically severing the structural graph connectivity.
- **Standardized Matrix Expression:** Paradigm B perfectly satisfies both the discrete binary constraint and the topological loop frustration preservation rules. Consequently, Paradigm B is formally established as the standardized algebraic pruning rule, explicitly formulated as:

$$\mathbf{M}_{n+1}(i, j) \leftarrow \chi(\sigma^{-1}(i), \sigma^{-1}(j)) \cdot \mathbf{M}_{n+1}(i, j) + [1 - \chi(\sigma^{-1}(i), \sigma^{-1}(j))] \cdot 1$$

7. Statistical Mechanics Derivation: Thermodynamic Origin of the Zero-Interference Boundary

To isolate non-coherent topological states and eliminate any analytical degeneracy in probability behaviors, the absolute zero-interference plateau ($\text{sgn} = 0$) must be derived independently from the microstate partition function of Status-Relational Entropy (SRE) dynamics.

Theorem 5 (SRE Hamiltonian Phase Separation Theorem)

The local topological Hamiltonian $\mathcal{H}_{\text{local}}$ driven by the un-realized path interference polynomial $\tilde{\mathcal{E}}_{\text{local}}(v_f, v_m)$ under the valuation homomorphism consists strictly of three mutually disjoint phase spaces:

$$\mathcal{H}_{\text{local}} = \begin{cases} -\mathcal{E}_{\text{local}} - 1, & \text{if } \tilde{\mathcal{E}}_{\text{local}} > 0 \text{ (Coherent Condensation Phase)} \\ 0, & \text{if } \tilde{\mathcal{E}}_{\text{local}} = 0 \text{ (Absolute Information Vacuum Phase)} \\ +\mathcal{E}_{\text{local}} + 1, & \text{if } \tilde{\mathcal{E}}_{\text{local}} < 0 \text{ (Frustrated Annihilation Phase)} \end{cases}$$

Proof

1. Based on the fundamental principles of SRE dynamics, the microscopic occurrence probability of a local front channel follows a canonical distribution governed by its local Hamiltonian, where the statistical weight evaluates to $\exp(-\beta \cdot \mathcal{H})$.
2. When $\tilde{\mathcal{E}}_{\text{local}} > 0$, the local loop is entirely free from topological frustration, allowing the network to condense coherently and lower its free energy. The Hamiltonian manifests as an attractive potential well, where the structural correction term inside the energy denominator yields $\exp(+1) = e$.
3. When $\tilde{\mathcal{E}}_{\text{local}} < 0$, the system plunges into deep structural frustration. The non-neutralizable phase conflict acts as an active gauge work. The Hamiltonian switches into a repulsive energy barrier, where the structural correction term scales down to $\exp(-\beta \cdot \Delta E_{\text{gauge}}) = \exp(-1) = e^{-1}$.
4. When $\tilde{\mathcal{E}}_{\text{local}} = 0$, a perfect algebraic cancellation occurs between positive and negative polarities across all 2-step paths. In the language of statistical physics, this local manifold possesses neither ordered-phase condensation nor phase-gauge work. The degeneracies of opposite-energy states are perfectly balanced, collapsing the local topological Hamiltonian strictly to zero: $\mathcal{H}_{\text{local}} \equiv 0$.
5. Substituting $\mathcal{H}_{\text{local}} = 0$ back into the Boltzmann canonical factor, the phase-separation correction term yields $\exp(0) = 1$. This eliminates any mathematical continuity gaps at the phase boundaries, ensuring complete decoupling between different topological states.

■ Theorem 5 is complete.

8. Endogenous Resolution: Grand Proof of the Global Absolute Vertex Degree Upper Bound K_0

Treating the maximum vertex-degree bound as an external hard-coded assumption introduces a systemic disconnect. We derive the absolute saturation boundary of the global-degree topology endogenously from the pruning-probability measure and SRE entropy changes.

Theorem 6 (Global Degree Saturation and Endogenous $O(1)$ Complexity Theorem)

For any newly injected frontier vertex v_f , its local first-order-neighborhood degree $\text{Degree}(v_f)$ asymptotically converges to an endogenously fixed global absolute constant upper bound K_0 as the full-network scale approaches the Thermodynamic Limit ($n \rightarrow \infty$). This upper bound is uniquely locked by the endogenous coupling constant and the causal slope.

Proof

1. Let the frontier incremental vertex v_f be injected into the network, attempting to establish connections with historical surviving vertices. By the definition of the reconstructed causal depth, as the evolutionary cascade scales up, the topological depth $\mathcal{D}_s = (n + 1) - \sigma(v_m)$ between v_f and any ancient surviving node v_m expands linearly without bound.
2. We examine the extreme scaling behavior of the system under large horizons. As $\mathcal{D}_s \rightarrow \infty$ while the coupling parameter remains bounded, we substitute the parameters into the adaptive-energy-level probability-mapping equation. The pruning probability that this channel drops into a dormant state satisfies:

$$\lim_{\mathcal{D}_s \rightarrow \infty} p(v_f, v_m) = 1 - \lim_{\mathcal{D}_s \rightarrow \infty} \frac{1}{1 + \lambda(n) \cdot \frac{\mathcal{D}_s}{\mathcal{E}_{\text{local}} + \exp(\text{sgn}(\tilde{\mathcal{E}}_{\text{local}}))}}$$

3. Because the local absolute frustration energy $\mathcal{E}_{\text{local}}$ is strictly bounded by local-neighborhood hash intersections, its maximum growth rate with respect to depth $\mathcal{D}_s$ has a partial derivative of exactly 0. Consequently, the internal incremental-fraction ratio satisfies:

$$\lim_{\mathcal{D}_s \rightarrow \infty} \frac{\mathcal{D}_s}{\mathcal{E}_{\text{local}} + \exp(\text{sgn}(\tilde{\mathcal{E}}_{\text{local}}))} = +\infty$$

4. Directly computing the limit yields:

$$p(v_f, v_m) = 1 - \frac{1}{1 + \infty} = 1 - 0 = 1.$$

5. This limit proves that the probability for ancient nodes to maintain non-zero functional connections with the current frontier **strictly and deterministically converges to 1** under massive multi-round scaling (i.e., a 100 % pruning / dormancy rate).
6. We define the effective attraction radius (coherence length) as ξ . Once causal-depth steps cross this critical threshold ($\mathcal{D}_s > \xi$), the binary random-decision gate collapses to dormant state $\chi \equiv 0$ with probability one.
7. According to Paradigm B (Elimination-Conduction Mechanism) selected in Section 6, these edges are all forced to constant +1, cutting off information flow in multiplicative feedback loops.
8. Accordingly, the number of historical vertices for which v_f can maintain non-unit spin and carry them into next-round multiplicative evolution is rigidly confined within finite generational window: $\sigma(v_m) \in [n + 1 - \xi, n + 1]$.
9. Set absolute constant upper bound $K_0 = \lceil \xi \rceil$. Since ξ is uniquely determined by statistical-mechanical inverse temperature β and fundamental constant λ_0 , it is algebraically independent of total system scale n . Hence:

$$\sup_{n \in \mathbb{N}^+} \max_{v \in V_n} \text{Degree}(v) \leq K_0 \quad \text{holds unconditionally.}$$

10. Substitute the endogenously-derived bound K_0 back into Theorem 2. The constancy of single-step elementary-algebraic-operation count $\mathcal{A}(n) \leq 3K_0^2$ follows naturally. The local-time-stream $O(1)$ complexity bound of the Second Operator receives its complete endogenous justification.

■ Theorem 6 is complete.

9. P0 Emergency Revision: Consistency Proof for Cycle-Space Frustration Criteria under Paradigm-B Pruning

Paradigm B uses forced assignment to $+1$ instead of adjacency zeroing. It must be proven that this operation does not irreversibly destroy frustration criteria or lose topological information for historical generalized-cycle-space bases.

Theorem 7 (Cycle-Space Frustration Homomorphism-Conservation Theorem under Paradigm-B Pruning)

Let $\mathcal{C}(G_n)$ denote the generalized cycle space of graph G_n , spanned jointly by fundamental cycles and vertex self-loops. After Second-Operator Paradigm-B pruning forces edge $e^* = (u, v)$ to spin-value $+1$, producing new graph G'_n . For any loop $L \in \mathcal{C}(G'_n)$ that does **not** contain edge e^* , its frustration invariant under evaluation homomorphism Φ is strictly conserved:

$$\prod_{e \in L} \Phi(S_n(e)) \equiv \prod_{e \in L} \Phi(S'_n(e)).$$

Proof

1. Algebraic essence of Paradigm-B pruning is a local selective-replacement homomorphism: $S'_n(e) = 1$ iff $e = e^*$, otherwise $S'_n(e) = S_n(e)$.
2. Consider arbitrary loop L belonging to new graph G'_n , with premise $e^* \notin L$.
3. Evaluate product over edges of L for modified graph:

$$\prod_{e \in L} \Phi(S'_n(e)) = \prod_{\substack{e \in L \\ e \neq e^*}} \Phi(S_n(e)).$$

4. Since e^* is strictly not an element of edge-set L , the set $\{e \in L \mid e \neq e^*\}$ is identical to original edge-set L .
5. Therefore the product reduces identically to original-graph product:

$$\prod_{\substack{e \in L \\ e \neq e^*}} \Phi(S_n(e)) \equiv \prod_{e \in L} \Phi(S_n(e)).$$

6. Frustration judgment for loop L depends entirely on sign of this product (-1 means frustrated; $+1$ frustration-free). Since product value remains unchanged, topological-frustration invariants are preserved and no irreversible loss of historical topological information occurs.

■ Theorem 7 is complete.

Corollary 7.1 (Quantitative Superposition Criterion for Arbitrary Historical Loops)

For historical local graphs free of frontier edges, given multiple overlapping fundamental loops $\{L_\alpha\}$. Define global multi-subgraph total-frustration superposition index Θ_{global} as weighted logarithmic sum of frustration eigenvalues for all loops. Avalanche-like cascaded pruning will be triggered on mesoscopic scale if and only if this index crosses critical free-energy threshold.

10. P0 Emergency Revision: Resolving Circular Dependency of $\lambda(n) - \rho(\mathbf{A}_n)$ and Reconstructing Coupling Constant

In earlier formulations, $\lambda(n)$ depended on spectral radius $\rho(\mathbf{A}_n)$ of current graph, while $\mathbf{A}_n$ itself was generated by pruning masks using $\lambda(n)$. This creates logical **non-linear circular dependency**. We eliminate the dependency by adopting prior spectral invariants from previous iteration and prove unique existence of analytic solution.

Theorem 8 (Decoupled Normalization-Constant Unique-Determinism Theorem)

Reconstruct explicit mapping for coupling-constant $\lambda(n)$ in terms of deterministic historical subgraph spectral radius $\rho(\mathbf{A}_{n-1})$ obtained from previous evolutionary round:

$$\lambda(n) = \frac{1}{\beta} \cdot \frac{\ln(1 + \rho(\mathbf{A}_{n-1}))}{n + 1}.$$

This reconstructed formula yields uniquely-determined real analytic solution at every frontier-expansion step and fully removes cross-operator circular dead-lock.

Proof

1. At start of n -th dimensional expansion step ($n \rightarrow n + 1$), historical subgraph $\mathbf{A}_{n-1}$ has already finished all pruning and evaluation mappings from previous iteration; hence $\mathbf{A}_{n-1}$ is a fully-deterministic real-valued sparse matrix.
2. By Perron-Frobenius theorem, spectral radius (maximum absolute eigenvalue) $\rho(\mathbf{A}_{n-1})$ of any real-symmetric sparse matrix is an unconditionally unique deterministic algebraic topological invariant.
3. Substitute unique scalar $\rho(\mathbf{A}_{n-1})$, pre-evolution step index n , macroscopic inverse-temperature constant $\beta > 0$ directly into reconstructed explicit formula.
4. Logarithm and division are single-valued monotonic functions over positive real domain $(0, +\infty)$. Right-hand side contains no unresolved future-dependent pruning-mask parameters.
5. Consequently $\lambda(n)$ acts purely as forward-looking prior control-parameter, its analytic solution is uniquely determined; circular-dependency conflict is fundamentally eliminated.

■ Theorem 8 is complete.

11. P0 Emergency Revision: Derivation of Explicit Topological-Weight w_α and its SRE Statistical-Mechanical Origin

Theorem 9 (Explicit Mapping of Topological-Weight w_α and Minimal-Action Derivation Theorem)

Within highly-overlapping multi-loop polygonal networks, explicit topological weight for each independent fundamental-cycle basis element L_α is defined as reciprocal of loop length: $w_\alpha = \frac{1}{|L_\alpha|}$. Its value range is strictly confined to interval $(0, \frac{1}{3}]$, and its thermodynamic origin strictly follows minimal-action principle of SRE path integral.

Proof

1. According to local causality principle of Status-Relational Entropy dynamics, phase loss for spin information propagating along long paths is proportional to topological step-count (edge-count $|L_\alpha|$) of that path.
2. Total local action exerted on frontier is superposition of local-path-integral contributions from all overlapping fundamental historical cycles: $\mathcal{S}_{\text{total}} = \sum_\alpha \mathcal{S}(L_\alpha)$.
3. To satisfy macroscopic-microscopic interaction minimal-action principle ($\delta\mathcal{S} = 0$), long loops suffer harmonic-series decay of coherent control strength upon frontier edges proportional to loop length. Hence contribution weight must be reciprocal of loop length:

$$w_\alpha = \frac{1}{|L_\alpha|}.$$

4. Boundary analysis: According to algebraic graph theory, shortest non-trivial closed loop is triangle satisfying $|L_\alpha| \geq 3$. Weight attains maximum value $w_\alpha = \frac{1}{3}$ for triangular loops. As long-loop length tends to thermodynamic limit $|L_\alpha| \rightarrow \infty$, weight converges to zero. Value range is strictly $(0, \frac{1}{3}]$.

■ Theorem 9 is complete.

12. P1 Supplementary Improvement: Symbolic-Polynomial-Domain Prospective Frustration-Judgement Theorem

It is necessary to establish prospective frustration-prediction mechanism directly on symbolic-polynomial ring $\mathcal{R}_\infty$, for performing formal-algebra pre-judgement before downstream evaluation homomorphism Φ is applied.

Theorem 10 (Prospective Frustration-Judgement Theorem over Symbolic-Polynomial Domain)

Define symbolic-domain independent two-step formal-interference polynomial $\mathcal{I}_{\text{sym}}(v_f, v_m) \in \mathcal{R}_\infty$. If this multivariate polynomial contains mutually-locking monomial combinations with even-order monomials and odd-order monomials of opposite signs, this channel can be pre-judged as **structurally-determined frustrated state**, before downstream numerical assignment.

Proof

1. Expand formal-interference term within multivariate-polynomial ring $\mathcal{R}_\infty = \mathbb{R}[\bigcup \mathcal{V}_i]$:

$$\mathcal{J}_{\text{sym}}(v_f, v_m) = \sum_k x_k \cdot y_k + \dots$$

2. Suppose polynomial algebraic structure contains irreducible inter-locked monomial combinations, for instance sub-structure like $x_1^2 - x_1 x_2$.
3. Apply arbitrary valid global-binary-assignment homomorphism Φ . Under binary-spin constraints, even-power monomials collapse to constant $+1$: $\Phi(x_1^2) \equiv 1$.
4. Odd-power monomials such as $\Phi(x_1 x_2)$ fluctuate within set $\{+1, -1\}$ depending on concrete spin assignments.
5. Even-order constant term $+1$ can algebraically cancel against fluctuating odd-order terms ($1 - 1 = 0$). This triggers non-linear jump of Hamiltonian towards repulsive-barrier regime $\mathcal{H}_{\text{local}} > 0$.
6. This energy-level collapse is purely determined by topological-locked structure inside symbolic matrix; prospective frustration-lock judgement can therefore be performed already in polynomial-input domain.

■ Theorem 10 is complete.

Explicit Algebraic Transformation for Third-Order Graph-Laplacian under Mask χ and Null-Space Compatibility Proof

After Boolean mask χ from Second-Operator drives Paradigm-B algebraic transformation on graph, resulting third-order graph Laplacian $\mathbf{L}^{(3)'}$ maintains strict isomorphism compatibility with null-space of original first-order operator and zero-order Betti-number $\beta_0 = 1$ topological-protection mechanism.

Proof

1. Paradigm-B pruning explicit matrix formula has been defined in Section 6.2:

$$\mathbf{M}_{n+1}(i, j) \leftarrow \chi \cdot \mathbf{M}_{n+1} + (1 - \chi) \cdot 1.$$

2. According to algebraic graph theory, construction of third-order graph Laplacian $\mathbf{L}^{(3)}$ is jointly built upon self-products of adjacency matrices and higher-order spectral projections; dimension of its null-space directly determines topological invariants of local branches.
3. Paradigm-B forces spin to value $+1$ without erasing adjacency-graph edges; degree matrix $\mathbf{D}' \equiv \mathbf{D}$ remains rigidly conserved.
4. Examine null-space property of first-order Laplacian: for all-ones vector $\mathbf{1}$, identity holds: $\mathbf{L}\mathbf{1} = (\mathbf{D} - \mathbf{A})\mathbf{1} = \mathbf{0}$.
5. After Paradigm-B transformation, modified adjacency matrix satisfies $\mathbf{A}'(i, j) \in \{+1, -1\}$ with no destructive zero entries breaking row-sum balance. Hence identity still holds:

$$\mathbf{L}'\mathbf{1} = (\mathbf{D}' - \mathbf{A}')\mathbf{1} \equiv \mathbf{0}.$$

6. This proves all-ones base vector permanently resides inside null-space $\text{Null}(\mathbf{L}^{(1)'})$. First-order null-space suffers no algebraic fracture; zero-order Betti-number is locked unconditionally to physical

requirement $\beta_0 = 1$ (single-connected-domain protection). Downstream third-order-Laplacian $\mathbf{L}^{(3)'$ null-space alignment-checks pass seamlessly.

■ Proof is complete.

13. Asymptotic Extremes and Boundary Analysis

13.1 Coupling Constant $\lambda \rightarrow 0$ (Zero-Dissipation Crystallization Boundary)

$$\lim_{\lambda \rightarrow 0} p(v_f, v_m) = 1 - \frac{1}{1 + 0} = 0 \implies \chi \equiv 1$$

- **Physical Interpretation:** The channel dormancy probability collapses strictly to zero, rendering the discrete gate a tautological true activation. The network loses macroscopic self-organizing dissipation and pruning-reduction capability; information propagates with unobstructed full long-range coherent condensation across all higher-order long-range chains. Topologically the network degenerates into deterministic globally fully-connected geometry (strong crystalline phase), with spatial-dimension dispersing towards infinity.

13.2 Coupling Constant $\lambda \rightarrow \infty$ (Infinite-Dissipation Heat-Death Boundary)

$$\lim_{\lambda \rightarrow \infty} p(v_f, v_m) = 1 - \frac{1}{1 + \infty} = 1 \implies \chi \equiv 0$$

- **Physical Interpretation:** Channel dormancy probability converges unconditionally to 100 %. Under Paradigm-B rules, all connected edges of topological network are forced algebraically to constant 1 upon first generation. System loses all capacity for polarity inversion and topological bifurcation; network spontaneously collapses into completely isotropic discrete isolated-point cluster without information interaction (Absolute-Vacuum phase); macroscopic heat-death is realized instantaneously.